



Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 23 Further Statistics 1

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General Comments

Overall, this paper was accessible to learners, and time did not appear to be a limiting factor. Whilst the paper contained three relatively long questions, each question was carefully scaffolded and structured to give learners opportunities to reset as they moved through the paper.

Question 1 provided an accessible introduction, and many learners were able to engage with the paper successfully. Question 2 represented a similar level of demand, and responses generally showed secure engagement with the question. Question 3 began with familiar and accessible processes before becoming more demanding in the final parts.

As the more sizeable and demanding parts appeared towards the end of the paper, learners had opportunities throughout the paper to demonstrate their knowledge and skills. The main discriminating features were the final two parts, the accuracy of calculations, and the quality of written responses.

Learners should be reminded to use the context of the question when writing hypotheses, conclusions to hypothesis tests, and comments on the limitations or assumptions of models. Some responses did not include the required key words or gave imprecise written responses.

Learners should also be reminded that questions may draw on knowledge and skills from both the AS Mathematics specification and the AS Further Mathematics specification. For example, in Question 2 learners were required to calculate a mean and a variance, and some responses did not apply these techniques accurately.

Finally, learners should be encouraged to support choices with clear reasoning. In part (d) of the final question, learners had to select either Sam or Chris and give a comparison or written reason for their choice. Some responses did not make this comparison explicit and therefore did not gain the final mark.

General Comments

Question 1

Parts (a) and (e)

These question parts went together to form a routine chi-squared test with a contingency table.

Most learners were able to give hypotheses in full context and gain the mark in (a). Though a number gave them “backwards”, with H_0 being ‘association’ and H_1 being ‘no association’.

In part (e), learners were required to complete the test using the given test statistic. Many responses gave the correct degrees of freedom and went on to make an appropriate conclusion. Some responses gave the critical value to 2 decimal places rather than the 3 decimal places shown in the tables, while others truncated rather than rounded the value. Learners should be reminded to use the values as given in the statistical tables where required and to give values to the appropriate degree of accuracy.

Part (b)

This part was one of the more challenging parts of the paper. Many responses recognised that cells should only be combined when the expected frequency is less than 5. However, fewer responses went on to calculate the expected frequency for the relevant cell, which was required for the second mark. Responses that calculated the expected frequency usually went on to provide sufficient working to gain both marks.

Some responses incorrectly stated that cells cannot be combined in contingency tables. These responses often did not go on to provide any further working, so did not gain either mark.

Parts (c) and (d)

These parts of the paper were very routine, and most responses were successful.

Part (f)

This part required learners to carry out a less familiar process. Many responses showed a correct calculation and gained both marks. Some responses completed the calculation correctly but did not explicitly state the minimum value of 36, which was required for the final mark.

Part (g)

This part required learners to apply their understanding of goodness-of-fit testing in a less familiar context. Many responses answered this part well. Marks were sometimes not gained where hypotheses were not stated accurately in context. Learners should be reminded to use the wording of the question to help them form their hypotheses and conclusions.

The final mark depended on learners using correct expected frequencies and a correct critical value. Some responses stated the critical value to too few decimal places, which meant that the final two marks could not be awarded.

Question 2

Part (a)

This part required learners to use the context of the question to state a condition or assumption for a Poisson model to be appropriate. Many responses used the context appropriately by referring to potholes and gained the mark. Some responses referred to the number of potholes rather than the rate of potholes, which was not sufficient for this mark.

Part (b)

In part (b), learners were required to calculate the mean and variance. Many responses completed these calculations correctly. Some responses did not use the correct formula for the variance, while others rounded values too early or to an inappropriate degree of accuracy, which affected the marks that could be awarded.

Part (c)

In part (c), learners needed to recognise that their values from part (b) were similar and state that this was consistent with the use of a Poisson distribution. Some responses did not give sufficiently accurate values for the mean and variance in part (b), but these values were often still close enough for learners to make a valid comparison and gain this mark.

Parts (d) and (e)

Part (d) was almost always answered correctly. Part (e) was also very well answered.

Part (f)

This part was generally answered well, with many responses giving suitable hypotheses. Some responses used $Po(5)$ rather than $Po(20)$, which meant that several marks could not be awarded.

Some responses used the correct distribution but found $P(Y = 28)$, which meant that only 2 the marks could be awarded of the 4 marks.

A few responses stated that the ‘number of potholes per kilometre has not increased’, which was not sufficient because the conclusion needed to refer to the **rate** of potholes. Learners should be reminded that the wording of the question can help them write conclusions in the required context. In this case, the **rate** of potholes occurring needed to be considered within each kilometre, rather than simply referring to the “number of potholes per kilometre”.

Question 3

Part (a)

Question 3 began with routine processes being required in part (a). As such learners responded well.

Part (b)

In part (b), learners were required to form and rearrange an equation. Many responses began with a correct equation, but some responses rearranged this equation incorrectly, so the second mark could not be awarded.

Part (c)

This was one of the challenging parts of the paper. Many responses did not identify the required approach. Most responses gained the first mark by giving outcomes for C and S , but some then considered independent values of X , using X_1 and X_2 rather than a single observation of X for both C and S .

A special case was considered for this approach, as these responses often included valid calculations that demonstrated relevant skills. However, as the approach itself was not correct, this special case could only gain 2 of the 4 marks.

Some responses calculated the expectations of C and S . This was not sufficient for marks in part (c) but could gain two marks in part (d) if carried out correctly.

Responses that used the same observation of X for both C and S generally went on to gain all 4 marks.

Part (d)

This final part was generally answered well. Many responses calculated $E(S)$ and then went on to provide sufficient working to gain all three marks. Some responses calculated $E(S)$ but did not explicitly compare it with $E(C)$, which was required for the final mark.